



LUCKY LIPS

Modern technologies are enabling communication without actually talking. Soon you could speak to another person without actually talking!

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What's been holding up the development of speech recognition? Common sense, for one. Imagine our crowded office environments with everyone around you screaming commands at their PCs. Or imagine yourself on a noisy street trying to get your stupid phone to call home, but only getting "please move to a quieter area" messages. Now imagine a technology that could allow everyone to make phone calls, and talk, without making a sound... serene, peaceful and heavenly, right? This year at CeBit, on of the largest trade fair showcasing digital IT and telecommunications solutions for home and work environments; a new concept called "Silent sound Technology" was demonstrated.

As the name suggests, it's a technology that helps you to transmit information without using your vocal cords. Cool isn't it? This technology aims to notice lip movements and transform them into a computer-generated sound that can be transmitted over a phone. Hence the person on the other end of phone receives the information in audio, as he would normally have. Scientists and researches at Karlsruhe Institute of Technology(KIT), Germany are working on this technology and claim this is the future of telecommunications.

The underlying technology When we generally speak aloud, air passes through the larynx and the tongue. Words are produced using articulator muscles in the mouth and the jaw region. Recently, it's proved that the articulator muscles become active irrespective of whether air

passes through them or not. Even by saying words without producing sound, weak electric currents are sent from your brain to the speech muscles. These electrical signal are known as electromyograms and this technique for evaluating and recording the electrical activity produced by skeletal muscles in general is known as Electromyography.

There is a device/sensor developed to work out the "Silent Sound Technology" at KIT. It monitors tiny muscular movements that occur when we speak by using electromyography and converts them into electrical pulses that can then be turned into speech - which simply means sound is produced without a word spoken.

The electromyographic sensor that is attached to the face, records the electrical signals produced by the facial muscles, compare them pre-recorded